



UNIVERSITY OF AMSTERDAM

Tutorial 1:

Practical Introduction Data Analysis

Variable/Scale Types, Regression Models & Probabilistic Attribution

Digital Marketing and Analytics: Week 1 Tutorial 1

Academic Year 2025-2026 – Semester 1 Block 2

AMSTERDAM
BUSINESS
SCHOOL

Business

Basics: Google Analytics

Google Analytics: Click Stream (Behavioral) Data

Google Analytics is used to review/analyse **click-stream online (behavioral)** data on **WEBSITE TRAFFIC**.

- Who visited our website (or clicked our content)? When ? How ?
- What kind of touchpoints are preceding this visit (online customer journey) Where are our visitors (from media sources) coming from ?



GA-4 (Basic Analytics): Aggregate Data total number of visitor and clicks

Example: How many people visited our website today-yesterday in total? How many (in total) coming from our Youtube Ad, how many from Google search?

GA-4 Big Query (Advanced Analytics): individual level data, so you can see each visitors' individual journey and visiting behavior.

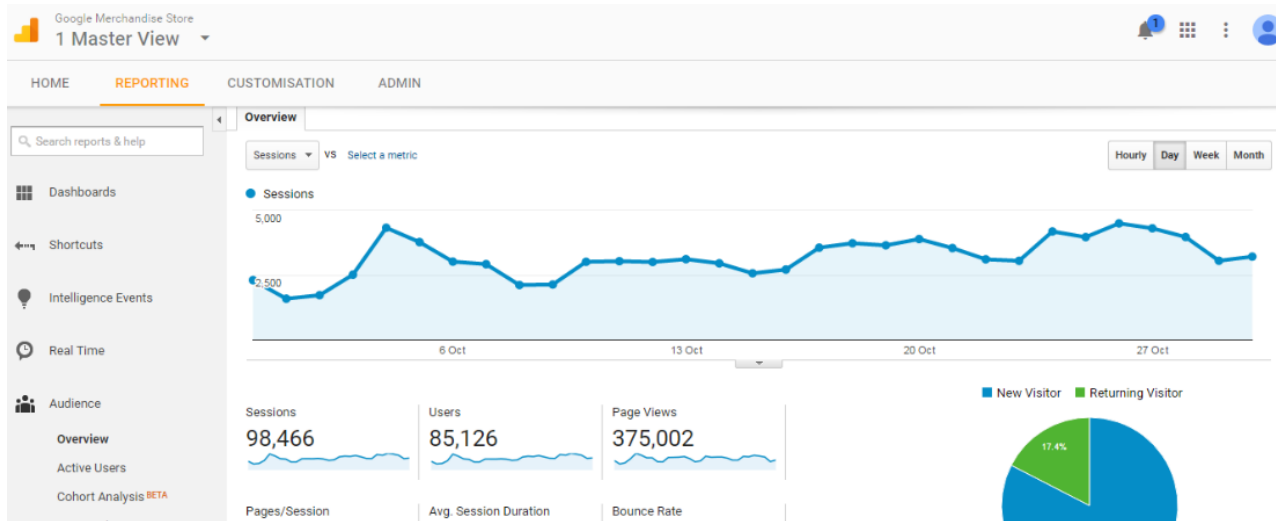
Example: What kind of online journey Customer A (John X) had today? - How was his click path? How about Customer B (Jane Y) – Did she see our Youtube AD?

Google Analytics (GA-4): Basics

On Google Analytics main-dashboard you can track the profile and behavior of your customers and visitors and what day do **over time**.

You can track the **general traffic to your website** from different **devices, locations, occasions and time-slots**.

Easy to use / Self Explanatory: Because GA is designed-developed for everyday use for managers at small to medium sized companies.



Google Analytics 4 (GA-4): Basis Platform – Online (Web) Analytics

GA-4 is the new generation analytics dashboard of Google for web-online analytics.

Please go to the Google Analytics 4 dashboard by using link and logging in with your Google (gmail) credentials.

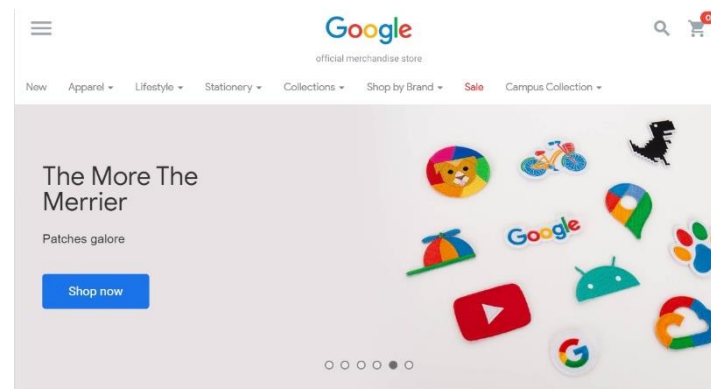
<https://analytics.google.com/>

This will connect you to the (demo) dashboard of Google Merchandise Store.

Create an Analytics account

Your first step is to set up an Analytics account, unless you already have one. Skip to [creating a property](#) unless you want to create a separate account for this website and/or app. For example, you might want to create another account if this website and/or app belong(s) to a separate business.

1. Go to <https://analytics.google.com> .
2. In [Admin](#) , click **Create**, then select **Account**.
3. Provide an account name. Configure the [data-sharing settings](#) to control which data you share with Google.
4. Click **Next** to add the first property to the account.

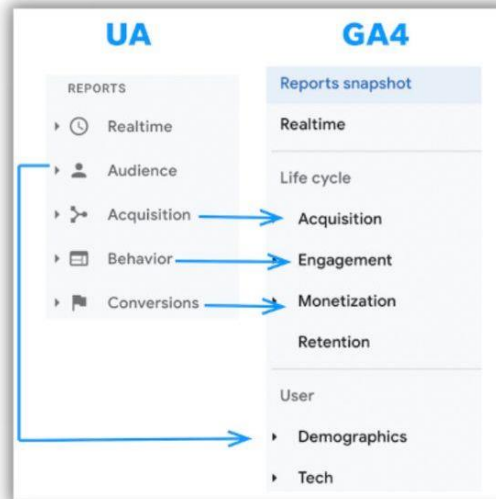


Google Analytics 4 (GA-4): What is New & Different

There is major improvements while Google Analytics evolves towards GA-4 new generation web analytics.

These updates are to better address the disruptive changes in **GDPR, Data Privacy-Protection regulations**, need for understanding other **EVENTS rather than conversions**, and new **granular-predictive nature of analytics**.

这些更新旨在更好地应对 GDPR、数据隐私保护法规带来的颠覆性变化，满足用户了解除转化之外的其他事件的需求，以及适应分析工具日益精细化和预测性的特点。



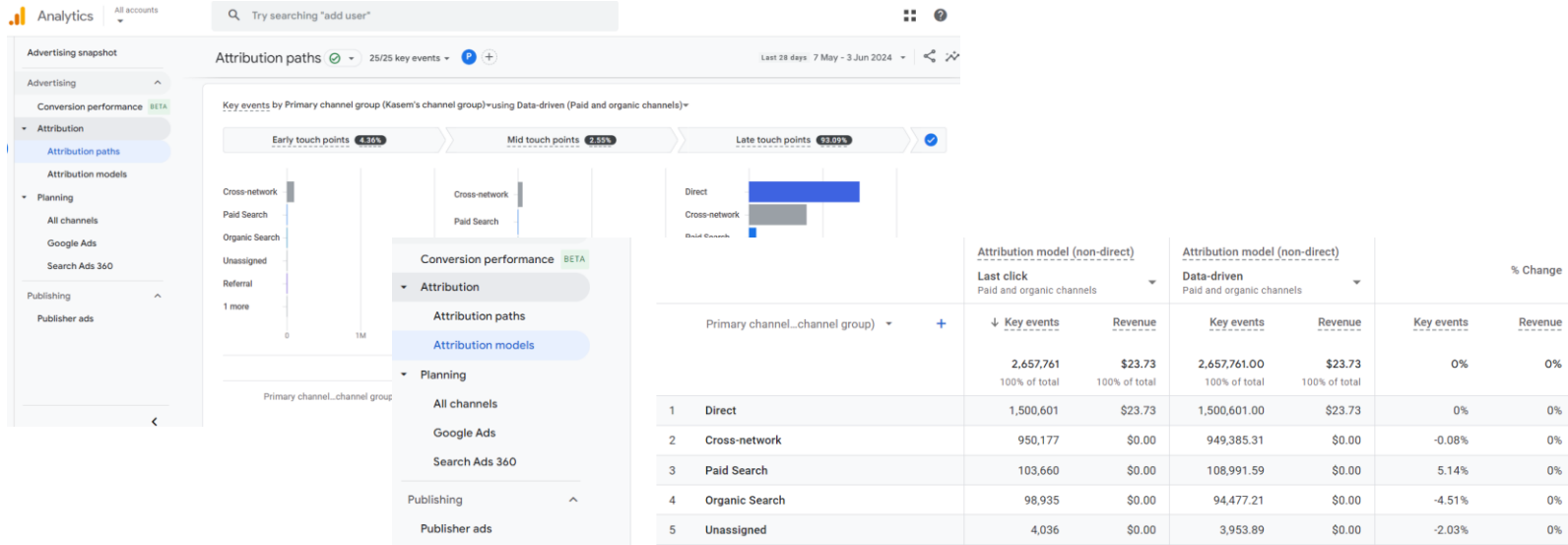
Google Analytics	Google Analytics
 Relies on tracking cookies which are increasingly limited.	 Less dependent on cookies and more privacy controls.
 Records sequences of sessions and pageviews.	 Records every interaction as an event to enable user-centric measurement.
 Potential gaps in user data from privacy measures.	 Predictive metrics and machine learning to fill in data gaps.
 Basic reporting and data visualization options.	 Advanced analysis tools previously only available in GA360.

Google Analytics 4 (GA-4): Web Analytics- Attribution

Attribution Module is available (also for beginners) by using Google Merchandise Store data-dashboard to have a taste of the platform.

GA-4 allows us to use various methods to allocate credits (value) to touchpoints.

In the example below Last-Click Model, and alternatively Data Driven attribution models are used.



Google Analytics 4 (GA-4): Micro-Conversions/Events in Web Analytics

Through **Google BQ-Attribution** it is also possible to define/track:

Micro actions (micro conversions) other than purchase – along with different features and touchpoint exposures (clicks-sessions) which provides even a more granular view of Customer Journey:

- Image impressions;
- Image clicks;
- Checking size;
- Selecting size;
- Reading more details;
- Viewed size map;
- Adding to Wishlist;
- Adding to cart, etc.

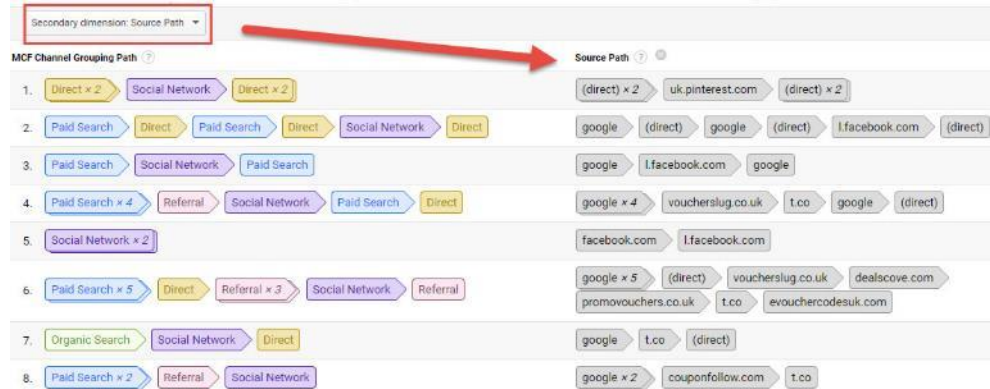


Google Analytics (GA-4): Practice Further

Google Analytics will **NOT** be learned in this course in depth:

- GA is basic-automated-plain tabulation on website traffic numbers-paths – no academic/ high-level course needed.
- There is no statistics, coding, economics background needed to use GA. Developed for every-day use of practitioners-managers (very self-explanatory)

Channel Grouping Path		Conversions
1.	Direct → Direct	32
2.	Non Branded Term → Non Branded Term	22
3.	Branded Term → Branded Term	14
4.	Referral → Referral	13
5.	Non Branded Term → Direct	13
6.	Non Branded Term → Non Branded Term → Non Branded Term	11
7.	Direct → Direct → Direct	7
8.	Branded Term → Direct	7
9.	Referral → Direct	6
10.	Referral → Non Branded Term	5



Pre-Analysis: Variable Types

Why This Intro to Variable-Scale Types and Regression

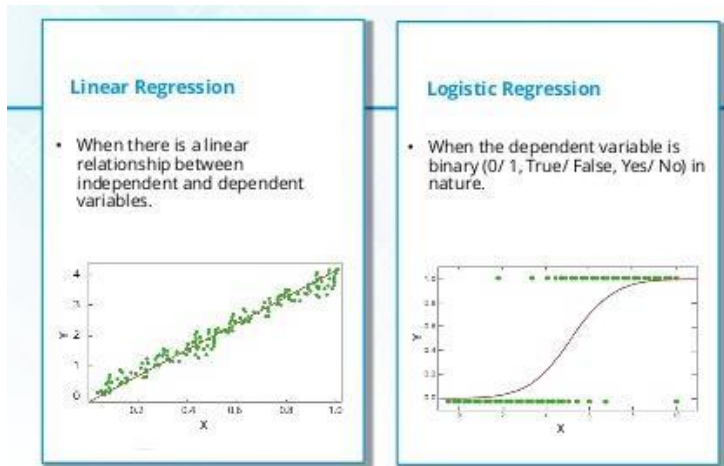
Some of you may have followed (MRM) 1-2 or QDA courses as a good basis for this course.

However: many others in this class did not – who didn't ?

*If I had the power I would have made it compulsory for all master students to follow these courses in pre-master – alas I don't have that power.

However to be able to analyze data – you need a basis (a simple understanding) of **scale types and simple analytical techniques**.

Thus: Please stay with me this intro is intended to give this basis in a nutshell for those who didn't – for others it is a good memory refresher



Variable/Scale Types

"Always the basis or departure point of our decision on which analysis to use"

Which Analysis/Regression Type to Use ? = We have to know the SCALE type of our Y variable.

We want to analyze EFFECT of certain variables on an outcome variable: **For example:** the effect of mobile ad exposure on online purchases

To be able to decide on **WHICH regression type to use**, we first know the **TYPE of our OUTCOME variable (Y)**

Then we can decide whether we use (a) multiple linear OR (b) logistic regression. But HOW?

Nominal	Ordinal	Interval	Ratio
"Eye color"	"Level of satisfaction"	"Temperature"	"Height"
Named	Named	Named	Named
	Natural order	Natural order	Natural order
		Equal interval between variables	Equal interval between variables
			Has a "true zero" value, thus ratio between values can be calculated

	Nominal	Ordinal	Interval	Ratio
Categories	●	●	●	●
Rank order		●	●	●
Equal spacing			●	●
True zero				●

Variable/Scale Types – Basis to Start

Carefully check the differences between measure (scale) types below.

This is something you immediately need to spot/tell to be able to correctly analyze the data and decide our analysis type !

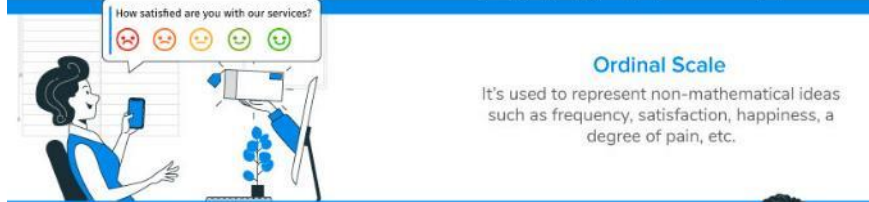
Nominal scale

It's used to label variables in different classifications and does not imply a quantitative value or order.



Ordinal Scale

It's used to represent non-mathematical ideas such as frequency, satisfaction, happiness, a degree of pain, etc.



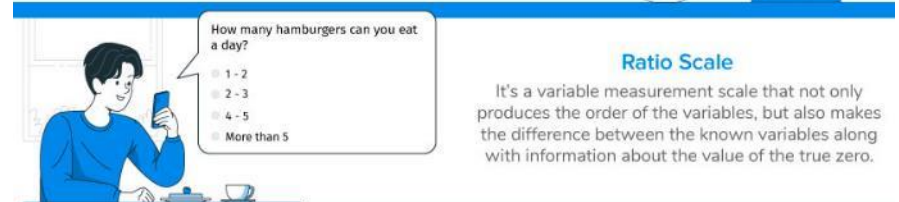
Interval Scale

It's defined as a numerical scale where the order of the variables as well as the difference between these variables is known.



Ratio Scale

It's a variable measurement scale that not only produces the order of the variables, but also makes the difference between the known variables along with information about the value of the true zero.



Variable/Scale Types > Regression Type

Then which regression to use: It is simpler than you may think.

To choose the correct regression type we look at the OUTCOME (Y) variable and its scale type.

*There are many other regression or analysis types that a user may consider using depending on: (a) the type of outcome variable (b) distribution of the data (c) linearity/non-linearity etc. However from a practical perspective (since this course is not a statistics or modeling course) we focus on **two simple and commonly used types** that we use a lot with online data.

(Y) IS METRIC/CONTINUOUS VARIABLE

TOTAL SALES PER PERSON (IN EURO)
TOTAL NUMBER OF CLICKS (#) OF AN ONLINE AD
TOTAL NUMBER OF UNITS SOLD
CHECK OUT: THEY ARE ALL REAL QUANTITIES!

MODEL-ANALYSIS:
MULTIPLE LINEAR REGRESSION

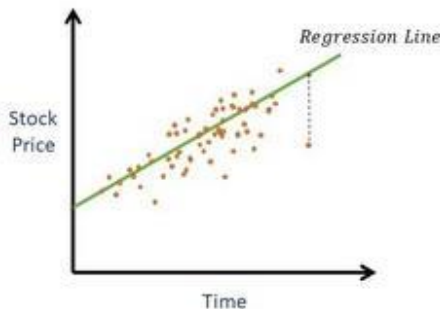
(Y) IS DISCRETE/NOMINAL VARIABLE

CONVERSION VS NO-CONVERSION (0/1)
PURCHASE VS. NO PURCHASE (0/1)
CLICKED VS. NOT CLICKED (0/1)
CHECK OUT: THEY ARE NOT QUANTITIES!
BUT MAINLY DIFFERENT SITUATIONS

MODEL-ANALYSIS: LOGISTIC REGRESSION

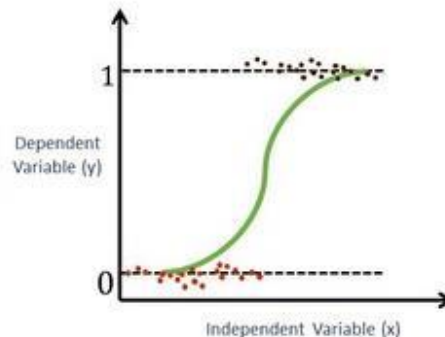
Linear Regression

- Aim is to predict continuous valued output.
- Output value can be any possible integer number.



Logistic Regression

- Aim is to predict the label for input data.
- Output is categorical (Binary) i.e. 0/1, True/False, etc.



Continuous Outcome (Y)

Linear Regression

Example: Drivers of Online Sales

Linear Regression in a Nutshell

$$Y = \alpha + \beta_1 X_1 + \beta_2 X_2 + \beta_i X_i + \varepsilon$$

- Y = dependent variable (DV) -- or also named as outcome variable (OV)
- X_i = independent variable (IV) – or also named as predictor variable (PV)
- α = constant : meaning = what happens when all (X) predictor variables are set to 0
- β_i = coefficient of PV i
- ε = residual/error term (part of Y not explained by collection of Xs in model)

Why use linear regression? Explaining the variance in continuous outcome (OV) with several other predictor variables (PVs)

- > So multiple predictor variables can be tested simultaneously
- > Identify the relative importance (effect) of each predictor variable (X) on outcome (Y)
- > Predict the values of the outcome variable (in our case: sales) from values of multiple predictor variable

When use multiple linear regression?

OV (what is the outcome?):

Quantitative/Metric Scale

PV (what are the potential drivers/predictors?):

One or more PVs, quantitative or categorical scale

Linear Regression Example: Drivers of Online Sales

Our Data: From an online retailer. Y= Total order amount (sales) per customer (in Euros) in one calendar year.

Our Question: What affects (drives) online purchases ? Which type of ads are more influential? Or is it the loyalty program ?

Our Predictors: Total number of clicks (#) for various online ads, and loyalty program membership status

*Online_Revenue_RegressionPractice_Data_v01.sav [DataSet1] - IBM SPSS Statistics Data Editor

	Facebook	BannerAds	SEA	Emails	LoyaltyProgram	TotalSalesEuro
1	0	2	0	0	0	80
2	0	2	1	1	1	100
3	3	2	4	5	2	230
4	0	2	1	2	1	120
5	0	2	1	2	1	110
6	5	4	4	1	2	400
7	1	4	2	4	0	100

*Online_Revenue_RegressionPractice_Data_v01.sav [DataSet1] - IBM SPSS Statistics Data Editor

	Name	Type	Width	Decimals	Label
1	Facebook	Numeric	1	0	# Facebook: number of Facebook ads clicked in 2017 (X1)
2	BannerAds	Numeric	1	0	# Banner Ads: number of Banner ads clicked in 2017 (X2)
3	SEA	Numeric	1	0	# SEA: number of Search Engine Advertising (Paid Search) results clicked in 2017 (X3)
4	Emails	Numeric	2	0	# Emails: number of Emails clicked/ opened in 2017 (X4)
5	LoyaltyProgram	Numeric	1	0	Loyalty Program Membership: 0=NO 1=Regular Member 2=Gold Member
6	TotalSalesEuro	Numeric	4	0	Total Sales per customer in Euros (Y= outcome or dependent variable)

Linear Regression Example: Drivers of Online Sales

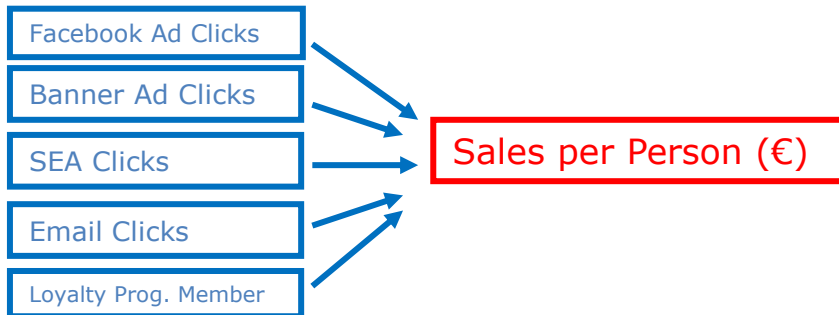
Research Question: Can we predict online sales (in Euro) by using ad-exposure (on different media) and loyalty program membership

Which potential driver is more influential on sales? Which ones do have a significant impact/ which ones do not?

	Name	T			Label
1	Facebook	...	1	0	# Facebook: number of Facebook ads clicked in 2017 (X1)
2	BannerAds	...	1	0	# Banner Ads: number of Banner ads clicked in 2017 (X2)
3	SEA	...	1	0	# SEA: number of Search Engine Advertising (Paid Search) results clicked in 2017 (X3)
4	Emails	...	2	0	# Emails: number of Emails clicked/ opened in 2017 (X4)
5	LoyaltyProgram	...	1	0	Loyalty Program Membership: 0=NO 1=Regular Member 2=Gold Member
6	TotalSalesEuro	...	4	0	Total Sales per customer in Euros (Y= outcome or dependent variable)

Predictor Variables (PVs)

Outcome Variable (OV)



Linear Regression Example: Drivers of Online Sales

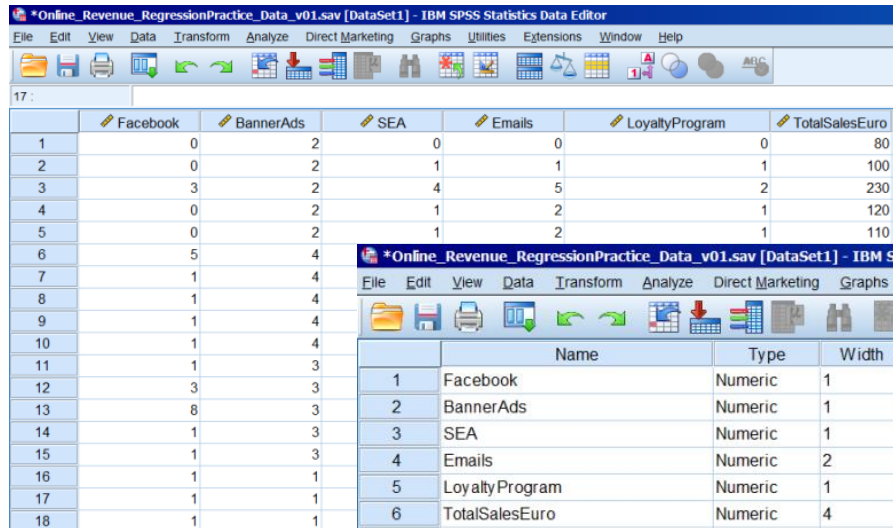
Total Sales in Euro (€) is a quantity-metric variable here > SO we use **LINEAR REGRESSION**

All **ad exposure variables** are also measured in numeric/metric scale: i.e **total number of times # each type clicked**.

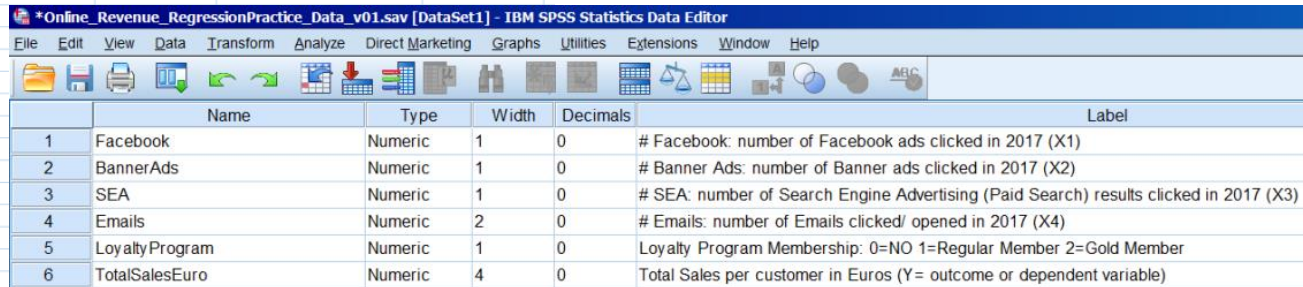
This means they can **directly be used in regression** (#facebook, #bannerads, #SEA, #emails)

Loyalty program variable however is a **discrete/categorical variable** (no-member = 0, regular = 1, gold member = 2)

Thus it has to be converted/recoded into dummy variables to be used in regression – we shall learn how to do it in next slides!



	Facebook	BannerAds	SEA	Emails	LoyaltyProgram	TotalSalesEuro
1	0	2	0	0	0	80
2	0	2	1	1	1	100
3	3	2	4	5	2	230
4	0	2	1	2	1	120
5	0	2	1	2	1	110
6	5	4				
7	1	4				
8	1	4				
9	1	4				
10	1	4				
11	1	3				
12	3	3				
13	8	3				
14	1	3				
15	1	3				
16	1	1				
17	1	1				
18	1	1				



	Name	Type	Width	Decimals	Label
1	Facebook	Numeric	1	0	# Facebook: number of Facebook ads clicked in 2017 (X1)
2	BannerAds	Numeric	1	0	# Banner Ads: number of Banner ads clicked in 2017 (X2)
3	SEA	Numeric	1	0	# SEA: number of Search Engine Advertising (Paid Search) results clicked in 2017 (X3)
4	Emails	Numeric	2	0	# Emails: number of Emails clicked/ opened in 2017 (X4)
5	LoyaltyProgram	Numeric	1	0	Loyalty Program Membership: 0=NO 1=Regular Member 2=Gold Member
6	TotalSalesEuro	Numeric	4	0	Total Sales per customer in Euros (Y= outcome or dependent variable)

Linear Regression Example: Drivers of Online Sales – Dummy Creation

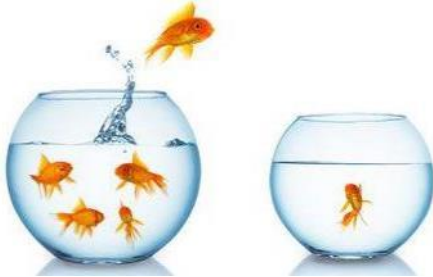
DUMMY (0-1) variables have to be created for our categorical (X) predictors – WHY?

Our regression results will tell us: What happens (how much increase we can expect in Y) if we increase our predictor by 1 unit $X > X+1$

HOWEVER –if we have a categorical variable changing status from a Loyalty Program NON-MEMBER to MEMBER > **Does NOT refer to an increase in quantity**

Thus we need to know what is the **differential effect of being a MEMBER vs. NON-MEMBER (status change)**

That means we first need to prepare them for the analysis as separate (DUMMY) variables:



VARIABLES

With five categories (c), you can create four (c – 1) dummy variables.
The omitted category (here WHITE) is known as the “referent” category.
Each of the four dummy categories is compared to the referent.

Original Variable Race-Ethnicity	Black	Hispanic	Asian	Other
White/European-American (1)	0	0	0	0
Black/African-American (2)	1	0	0	0
Hispanic/Latino (3)	0	1	0	0
Asian-American (4)	0	0	1	0
Other (5)	0	0	0	1

Let's recall:

3 categories in our Loyalty Program Variable

0= NO MEMBER

1= REGULAR

2= GOLD MEMBER

That means (according to the rule)

We shall create 2 (thus: 3-1) dummies:

NO MEMBER being the referent category

Categorical Variables (loyalty program membership) cannot be directly used in a regression model as a Predictor Variable (X)

They should first be converted into DUMMY VARIABLES to be used in the regression along with other X (numeric ones)

Linear Regression Example: Drivers of Online Sales – Dummy Creation

GENERAL RULE: NUMBER OF DUMMIES = NUMBER OF CATEGORIES – 1

One category being the basis (or reference) category

LOYALTY PROGRAM MEMBERSHIP has 3 categories: NO MEMBER, REGULAR AND GOLD MEMBER

NO MEMBER=0, REGULAR=1, GOLD MEMBER=2

For THREE categories: We need TWO dummies: **We call “NO MEMBER” the reference (basis) group;**

Coefficients for dummies express differences with this reference group. What happens when a person goes from being a NO-MEMBER to MEMBER

“REGULAR” dummy refers to difference in SALES between REGULAR and NO-MEMBER = in data REGULAR = 1 all others = 0

“GOLD” dummy refers to difference in SALES between GOLD and NO-MEMBER = in data GOLD = 1 all others = 0

	D1="REGULAR"	D2="GOLD"
NO MEMBER	0	0
REGULAR	1	0
GOLD	0	1

Linear Regression Example: Drivers of Online Sales – Dummy Creation

Make sure reference group has code 0, other group(s) have code 1 – for each dummy (see below)

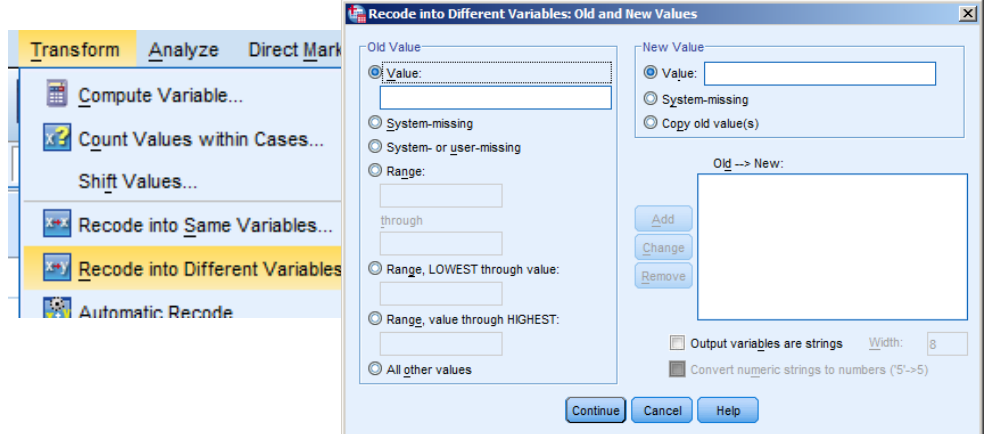
Some variables are naturally dummy if there are only two categories: men=0, women=1 > just pick one as the referent group (here: man)

With multiple dummies, recoding necessary: In SPSS >

Via Transform -> Recode > Into DIFFERENT variables (check in SPSS very easy)

Always keep original data intact! Create your dummies (0-1) in different variables.

	Original variable	Manually created dummies	
	LOYALTY PROGRAM	Dummy REGULAR	Dummy GOLD
NOMEMBER	0	0 -> 0	0 -> 0
REGULAR	1	1 -> 1	1 -> 0
GOLD	2	2 -> 0	2 -> 1



Linear Regression Example: Drivers of Online Sales

NOW our data is ready for analysis:

- We know our OUTCOME VARIABLE (Y): Total Sales (per person)
- We know our PREDICTOR VARIABLES (X): #facebook, #banner, #SEA, #emails and Loyalty Program Membership (see below)
- Finally we now have two DUMMY PREDICTORS (X-dummies) for LOYALTY PROGRAM membership: REGULAR and GOLD

	Facebook	BannerAds	SEA	Emails	LoyaltyProgram	REGULAR	GOLD	TotalSalesEuro
1	0	2	0	0	0	,00	,00	80
2	0	2	1	1	1	1,00	,00	100
3	3	2	4	5	2	,00	1,00	230
4	0	2	1	2	1	1,00	,00	120
5	0	2	1	2	1	1,00	,00	110
6	5	4	4	1	2	,00	1,00	400
7	1	4	2	4	0	,00	,00	100
8	1	4	0	4	0	,00	,00	90

SEE HOW DUMMIES
ARE CREATED/CODED
(IN RED)

	Name	T...	...	Label	Values	Missing	Columns	Align	Measure
1	Facebook	...	1	0 # Facebook: number of Facebook ads clicked in 2017 (X1)	None	None	12	Right	Scale
2	BannerAds	...	1	0 # Banner Ads: number of Banner ads clicked in 2017 (X2)	None	None	12	Right	Scale
3	SEA	...	1	0 # SEA: number of Search Engine Advertising (Paid Search) results clicked in 2017 (X3)	None	None	12	Right	Scale
4	Emails	...	2	0 # Emails: number of Emails clicked/ opened in 2017 (X4)	None	None	12	Right	Scale
5	LoyaltyProgram	...	1	0 Loyalty Program Membership: 0=NO 1=Regular Member 2=Gold Member	None	None	21	Right	Scale
6	REGULAR	...	8	2 Loyalty Dummy 1: Regular vs No-Member	None	None	24	Right	Nominal
7	GOLD	...	8	2 Loyalty Dummy 2: Golden vs No-Member	None	None	26	Right	Nominal
8	TotalSalesEuro	...	4	0 Total Sales per Customer in Euros (Y= outcome or dependent variable)	None	None	12	Right	Scale

Linear Regression Example: Drivers of Online Sales

NOW we can start with our regression analysis:

The screenshot shows the IBM SPSS Statistics Data Editor with a dataset named 'Online_Revenue_RegressionPractice_Data_v01.sav'. The 'Analyze' menu is open, and the 'Regression' option is selected. The 'Linear Regression' dialog box is also open, showing the 'Dependent' variable as 'Total Sales per customer in Euros (Y= outco...' and the 'Predictor Variables' as '# Facebook: number of Facebook ads click...', '# Banner Ads: number of Banner ads click...', '# SEA: number of Search Engine Advertisin...', '# Emails: number of Emails clicked/ opened...', 'Loyalty Dummy 1: Regular vs No-Member [...', and 'Loyalty Dummy 2: Golden vs No-Member [G...'. The 'Method' is set to 'Enter'. The 'Selection Variable' is empty. The 'Case Labels' and 'WLS Weight' fields are also empty. The 'OK' button is highlighted.

Outcome Variable

Predictor Variables

ALL VARIABLES IN DATA SET

ONLY DUMMIES FOR LOYALTY PROGRAM MEMBERSHIP

Linear Regression Example: Drivers of Online Sales

STEP 1: Understanding which X (predictor) variable has **significant** impact on Y

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	48,308	27,560		1,753	,084
	# Facebook: number of Facebook ads clicked in 2017 (X1)	17,863	4,847	,281	3,685	,000
	# Banner Ads: number of Banner ads clicked in 2017 (X2)	1,131	4,612	,017	,245	,807
	# SEA: number of Search Engine Advertising (Paid Search) results clicked in 2017 (X3)	,168	4,813	,002	,035	,972
	# Emails: number of Emails clicked/ opened in 2017 (X4)	2,513	4,160	,041	,604	,548
	Loyalty Dummy 1: Regular vs No-Member	62,924	22,584	,204	2,786	,007
	Loyalty Dummy 2: Golden vs No-Member	277,725	33,240	,693	8,355	,000

a. Dependent Variable: Total Sales per customer in Euros (Y= outcome or dependent variable)

Sig. has to be $< 0,05$ (95% confidence level) to indicate a **SIGNIFICANT EFFECT** ; therefore:

Facebook Ads (the # people clicked on facebook ads) and Loyalty Program Membership (both Regular and Gold) has a significant effect on Online Sales

Banner Ads, Search Engine Advertising and Email Marketing DO NOT seem to have a significant impact on sales; All (Sig) values $> 0,05$ (not significant)

Marketing Implications:

Facebook ADS and Loyalty Program seem useful in driving SALES. Thus: consider investing more on these touchpoints or prioritize them over the others !

*Banner Ads, SEA and Email Marketing DO NOT have an impact on creating more sales: we better reconsider our investment/use of these channels
OR; we should review/revise the way we use these channels: Is it the timing? Is it or is it the content (not right message) :*

Linear Regression Example: Drivers of Online Sales

STEP 2: Understanding/Interpreting effects (**unstandardized** beta coefficients)

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	48,308	27,560		1,753	,084
	# Facebook: number of Facebook ads clicked in 2017 (X1)	17,863	4,847	,281	3,685	,000
	# Banner Ads: number of Banner ads clicked in 2017 (X2)	1,131	4,612	,017	,245	,807
	# SEA: number of Search Engine Advertising (Paid Search) results clicked in 2017 (X3)	,168	4,813	,002	,035	,972
	# Emails: number of Emails clicked/ opened in 2017 (X4)	2,513	4,160	,041	,604	,548
	Loyalty Dummy 1: Regular vs No-Member	62,924	22,584	,204	2,786	,007
	Loyalty Dummy 2: Golden vs No-Member	277,725	33,240	,693	8,355	,000

a. Dependent Variable: Total Sales per customer in Euros (Y= outcome or dependent variable)

Continuous X predictor:
“What happens to Y if we increase X by one unit?”

Dummy X predictor:
“What is the impact on Y, if X changes status from reference to X_Dummy1 status?”

Unstandardized Beta coefficients help us to interpret the **absolute impact** of each touchpoint on sales; we interpret them like:

Facebook: One more (increase) click on our Facebook ads is likely to increase our sales per person 17,683 € per year.

Loyalty Program: Dummies are interpreted mostly in the same way (slightly differently)

Being a REGULAR member of Loyalty Program is likely to cause 62,924 € more sales in comparison to being a NON-MEMBER

Being a GOLD member of Loyalty Program is likely to cause 277,725 € more sales in comparison to being a NON-MEMBER

Linear Regression Example: Drivers of Online Sales

STEP 3: Comparing the effects; which X variable has a stronger impact on Y (**standardized beta** coefficients)

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	48,308	27,560		1,753	,084
	# Facebook: number of Facebook ads clicked in 2017 (X1)	17,863	4,847	,281	3,685	,000
	# Banner Ads: number of Banner ads clicked in 2017 (X2)	1,131	4,612	,017	,245	,807
	# SEA: number of Search Engine Advertising (Paid Search) results clicked in 2017 (X3)	,168	4,813	,002	,035	,972
	# Emails: number of Emails clicked/ opened in 2017 (X4)	2,513	4,160	,041	,604	,548
	Loyalty Dummy 1: Regular vs No-Member	62,924	22,584	,204	2,786	,007
	Loyalty Dummy 2: Golden vs No-Member	277,725	33,240	,693	8,355	,000

a. Dependent Variable: Total Sales per customer in Euros (Y= outcome or dependent variable)

Variables are measured in different scales; thus Unstandardized Coefficients cannot be used to compare effects.

Standardization > to make them comparable (ads = #clicks ; loyalty=member/not) – and then we interpret through St. Beta coefficients:

Being a **GOLD member** of the Loyalty Program seems to have the **STRONGEST** impact on sales:

Having the highest standardized Beta Coefficient: 0,693 (GOLD) > 0,281 (FACEBOOK) > 0,204 (REGULAR)

Linear Regression: Drivers of Online Sales - Interactions

The Concept of **INTERACTION**: What is the **combined effect** of two different (X) variables?

In practical words: What happens if we use **two different touchpoints (marketing actions) together** for the same group of people ?

Examining the interaction effects between two variables is very easy:

- 1 You compute an interaction variable by simply multiplying two variables of interest: what happens Facebook is used for Gold Members ?
- 2 Place your new (interaction) variable – along with all others that you used before; and re-run your regression again !

The image displays two screenshots from the SPSS software interface, illustrating the process of creating and using an interaction variable.

Left Screenshot: Compute Variable

- Target Variable:** FacebookLoyaltyGold_Interaction
- Numeric Expression:** Facebook * Loyalty_Golden_Dummy2
- Type & Label:** A list of variables is shown, including Facebook, Banner Ads, SEA, Emails, Loyalty Program Membership, Loyalty Dummy 1, Loyalty Dummy 2, and Total Sales per customer in Euros.
- Calculator:** A numeric keypad is visible, used for entering the expression.

Right Screenshot: Linear Regression

- Dependent:** Total Sales per customer in Euros (Y= outcome or dependent v.)
- Block 1 of 1:** A list of independent variables is shown, including Facebook, Banner Ads, SEA, Emails, Loyalty Program Membership, Loyalty Dummy 1, Loyalty Dummy 2, and FacebookLoyaltyGold_Interaction.
- Method:** Enter
- Selection Variable:** (Empty)
- Case Labels:** (Empty)
- WLS Weight:** (Empty)

A large blue arrow points from the Compute Variable dialog to the Linear Regression dialog, indicating the flow of the process.

Linear Regression Example: Drivers of Online Sales

Now let's interpret our regression results this time with the interaction variable:

- 1 You see **FacebookXGoldMember** interaction variable has a very strong positive impact on sales !
- 2 We still have a strong main (stand-alone) effect of Gold Membership on sales
- 3 But you will also notice the main (stand-alone) effect of Facebook vanishes/disappears – when we have the new interaction variable/effect
- 4 This means that the effect of Facebook ads on online sales can only be pronounced when Facebook Ads are shown to Gold Loyalty Club members!

Marketing Conclusion: Use Facebook Ads mainly for Loyalty Club members with Gold status !

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
	B	Std. Error	Beta			
1	(Constant)	71,832	25,200		2,850	,006
	# Facebook: number of Facebook ads clicked in 2017 (X1)	6,915	4,972	,109	1,391	,168
	# Banner Ads: number of Banner ads clicked in 2017 (X2)	1,705	4,127	,025	,413	,681
	# SEA: number of Search Engine Advertising (Paid Search) results clicked in 2017 (X3)	-5,267	4,471	-,075	-1,178	,242
	# Emails: number of Emails clicked/ opened in 2017 (X4)	2,053	3,722	,034	,551	,583
	Loyalty Dummy 1: Regular vs No-Member	75,364	20,389	,245	3,696	,000
	Loyalty Dummy 2: Golden vs No-Member	137,604	43,056	,343	3,196	,002
	FacebookXLoyaltyGold_Interaction	44,423	9,873	,549	4,499	,000

a. Dependent Variable: Total Sales per customer in Euros (Y= outcome or dependent variable)

Categorical (0/1) Outcome (Y)

Logistic Regression

Example: Multitouchpoint Attribution

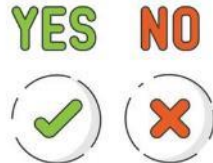
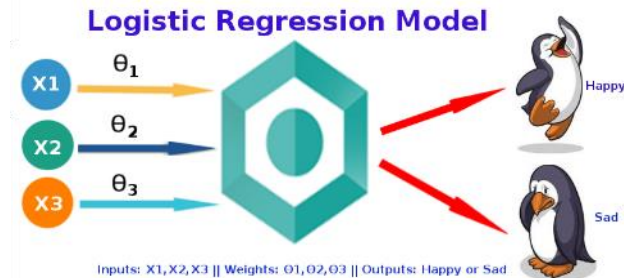
Logistic Regression Example: Multitouchpoint Attribution

This is to show you how model-based attribution works (remember Lecture 1) !

Pay Attention: NOW our outcome variable (Y) is NOT a quantity. Instead it is: categorical: **0/1 (Conversion vs. No Conversion)**

So we use LOGISTIC REGRESSION to see: Which TOUCHPOINTS are more INFLUENTIAL on conversion (online purchases) ?

**OUTCOME VARIABLE (Y):
CONVERSION ? – YES or NO**



**METHOD:
LOGISTIC REGRESSION**

Logistic Regression Example: Multitouchpoint Attribution

Let's say we have a small-data set on clicks and purchases and a simple (logistic) regression with results !

This is our data used in the model:

- 1 Clicks on each medium/touchpoint (whether or not clicked) X variables or predictors
- 2 Conversion (Purchase) in the end Y variable or outcome variable

	Ad1	Ad2	Youtube	Twitter_Link	Facebook_Link	Organic_Search_Result	Sponsored_Search_Result	Conversion
1	1	0	0	0	1	0	0	0
2	1	0	0	0	1	0	0	0
3	1	0	0	0	0	1	0	1
4	0	0	1	0	0	0	0	0
5	0	0	1	0	0	0	0	0
6	1	1	1	0	1	0	0	0
7	1	1	0	0	1	0	0	0
8	0	1	0	0	1	1	1	1
9	0	0	1	1	1	1	1	1
10	1	0	1	1	1	0	0	0
11	0	0	0	1	1	0	0	0
12	0	0	0	1	1	0	1	0
13	0	1	0	1	1	1	1	1

Touchpoints:
Whether or not customer
clicked each touchpoint
1=Clicked
0=NO click

IMPORTANT (Y)
Remember the difference between our OV
now and in the previous example
NOW we have a categorical : 0/1 OV
CONVERSION:
Whether or not customer purchased :
1=Purchased
0=NOT

Logistic Regression Example: Multitouchpoint Attribution

Next, we run (estimate) our ATTRIBUTION model (Logistic Regression)

Left hand side (Covariates/Affecting on..) : Touchpoint clicks = 0 or 1

Right hand side (Dependent/Is Affected) : Purchase (status) = 0 or 1

The screenshot shows the SPSS Data Editor with a dataset named 'Data_Attribution.sav'. The dataset has columns for touchpoints (Ad1, Ad2, Youtube, Twitter_Link, Facebook_Link, Organic_Search, Sponsored_Search) and a dependent variable (Conversion). A Logistic Regression dialog box is open, showing the dependent variable as 'Conversion' and the independent variables as 'Ad1(Cat)', 'Ad2(Cat)', 'Youtube(Cat)', 'Twitter_Link(Cat)', and 'Facebook_Link(Cat)'. The method is set to 'Enter'.

	Ad1	Ad2	Youtube	Twitter_Link	Facebook_Link	Organic_Search	Sponsored_Search	Conversion
1	1	0	0	0	0	1	0	0
2	1	0	0	0	0	1	0	0
3	1	0	0	0	0	1	0	0
4	1	0	0	0	0	1	0	0
5	1	0	0	0	0	1	0	0
6	1	0	0	0	0	1	0	0
7	1	0	0	0	0	1	0	0
8	1	0	0	0	0	1	0	0
9	1	0	0	0	0	1	0	0
10	1	0	0	0	0	1	0	0
11	1	0	0	0	0	1	0	0
12	1	0	0	0	0	1	0	0
13	1	0	0	0	0	1	0	0
14	1	0	0	0	0	1	0	0
15	1	0	0	0	0	1	0	0
16	1	0	0	0	0	1	0	0
17	1	0	0	0	0	1	0	0
18	1	0	0	0	0	1	0	0

Conversion (Y) OUTCOME
This is (likely) to be influenced by.... X variables

Touchpoints (X variables):
These are the touchpoints that can influence CONVERSION

Our KEY question in this data/project:
Which touchpoints are REALLY INFLUENTIAL on conversion

Logistic Regression Example: Multitouchpoint Attribution

Which touchpoints do have a (statistically significant) impact on conversion? Which touchpoint is the most influential?:

- i) Sig values as in multiple regression shows which are significant
- ii) **Exp(B)** – Odds Ratio shows us what is the most influential touchpoint

Variables in the Equation		B	S.E.	Wald	df	Sig.	Exp(B)
Step 1 ^a	Ad 1(1)	-1,320	,969	1,858	1	,173	,267
	Ad 2(1)	-,710	,999	,505	1	,477	,492
	Youtube(1)	1,100	,994	1,225	1	,268	3,004
	Twitter_Link(1)	,572	,976	,343	1	,559	,565
	Facebook_Link(1)	2,991	1,337	5,001	1	,025	19,898
	Organic_Search_Result (1)	5,901	1,423	17,196	1	,000	365,392
	Sponsored_Search_Result(1)	-,873	1,015	,740	1	,390	,418
	Constant	-2,858	1,374	4,326	1	,038	,057

a. Variable(s) entered on step 1: Ad 1, Ad 2, Youtube, Twitter_Link, Facebook_Link, Organic_Search_Result, Sponsored_Search_Result.

Most influential: Organic Search
Highest Exp(B) Showing the impact

Touchpoints:
FACEBOOK link
Organic Search

Significantly influential!
Both have effects with <0,05
(significant) significance levels
With high Exp(B) values!

Logistic Regression Example: Multitouchpoint Attribution

You don't need to know the theory (econometrics) or statistics behind this. This goes beyond the scope of this course. HOWEVER:

You NEED TO KNOW how to interpret the results of a simple attribution model (the table below)!

This is useful when you are manager (not doing the analysis – but knowing how to read, interpret it and draw the right conclusions) !

Organic Search with Exp(B) Odds Ratio 365,392 means:

Customers who clicked Organic Search results are **X 365 times more likely to convert (buy)** in comparison to those who did not click Organic Search Results

I know it is overly high & exaggerated (maybe not realistic) but I keep the effect HIGH for demonstration purposes – since this is an example :0)

Variables in the Equation						
		B	S.E.	Wald	df	Sig.
Step 1 ^a	Ad 1(1)	-1,320	,969	1,858	1	,173
	Ad 2(1)	-,710	,999	,505	1	,477
	Youtube(1)	1,100	,994	1,225	1	,268
	Twitter_Link(1)	-,572	,076	,242	1	,622
	Facebook_Link(1)	2,991	1,337	5,001	1	,025
	Organic_Search_Result(1)	5,901	1,423	17,196	1	,000
	Sponsored_Search_Result(1)	-,873	1,015	,740	1	,390
	Constant	-2,858	1,374	4,326	1	,038

Most influential: Organic Search
Highest Exp(B) Showing the impact

Touchpoints:
FACEBOOK link
Organic Search

Significantly influential!
Both have effects with <0,05
(significant) significance levels
With high Exp(B) values!

a. Variable(s) entered on step 1: Ad 1, Ad 2, Youtube, Twitter_Link, Facebook_Link, Organic_Search_Result, Sponsored_Search_Result.

Appendix: Regression with R / Python

Regression with R, Python or Other Software

ALL ANALYSIS that you can do with other software or analytical environments such as: R, PYTHON (or STATA)

Everything is the same as long as you get the logic behind – just the interface (the style you see it) and the scripts/codes are slightly different.

We use R or Python in most of our analytical (academic) work, however SPSS is handy for introductory and teaching purposes (at the beginner level), especially for those with no prior experience or background with data/data analysis.

You are **free to use whatever software in your assignments** – Their output tables are identical to each other (all includes Beta coefficients, P significance etc)

Features	Stata	SPSS	SAS	R & PYTHON
Learning curve	Steep/gradual	Gradual/flat	Pretty steep	Pretty steep
User interface	Programming/point-and-click	Mostly point-and-click	Programming	Programming
Data manipulation	Very strong	Moderate	Very strong	Very strong
Data analysis	Powerful	Powerful	Powerful/versatile	Powerful/versatile
Graphics	Very good	Very good	Good	Excellent

This Course:
Only for interpreting
regression outcomes

OK/ Voluntary if Preferred

Though not responsible in the exam
NOTE: as long as the data is same
they produce the SAME results
Whatever software you use !

Regression with Python – Pandas & Statsmodels



You may **not be** familiar with Python before this course. Regression with Python is **exactly same** as it is with SPSS or R:

- 1 If you know which regression type to use: Linear (OLS) or Logistic
- 2 If you know what your explanatory variables (X) and what your outcome variable (Y) is.

Command Prompt

```
Microsoft Windows [Version 10.0.18363.1801]
(c) 2019 Microsoft Corporation. All rights reserved.
```

```
H:\>pip install pandas_
```

```
Microsoft Windows [Version 10.0.18363.1801]
(c) 2019 Microsoft Corporation. All rights reserved.
```

```
H:\>pip install statsmodels
```

STEP 1

If you have not used Python before:

You always start by installing **pandas** and **statsmodels** libraries

This you do by using Command Prompt of your computer

pandas and **statsmodels** are the most commonly used and practical packages in python for data analysis & statistics etc.



To work with PYTHON conveniently we suggest to use a one of the Python development environments (editors) since it will be much easier to save, correct and take notes about what you have been doing.

*There are many editors-development environments however we recommend: **GOOGLE COLABS***

It is a cloud based Python development environment-editor powered by Jupyter – by using Colab you can easily access, edit, work and share your python projects on your Google Drive.

Regression with Python – Pandas & Statsmodels*



Then we use the same dataset (Online Sales) for **Multiple Linear Regression** as we used in SPSS/R. You will see we will have same results:

STEP 2 – Import the downloaded libraries for your analysis
STEP 3 - Import and read your data set (here from spss file)

```
Current working directory: C:\python
import pandas as pd
from sklearn import linear_model
import statsmodels.api as sm

df = pd.read_spss('A_Online_Sales_Regression_PracticeData.sav')
df.tail()
```

	Facebook	BannerAds	...	TotalSalesEuro	FacebookXLoyaltyGold_Interaction
79	0.0	7.0	...	10.0	0.0
80	0.0	7.0	...	75.0	0.0
81	1.0	7.0	...	80.0	0.0
82	1.0	7.0	...	160.0	0.0
83	3.0	0.0	...	210.0	0.0

STEP 4 – Now define your X (explanatory) and Y (Outcome) variables in your model

```
X = df[['Facebook','BannerAds','SEA','Emails','Loyalty_Regular_Dummy1','Loyalty_Golden_Dummy2']]
Y = df['TotalSalesEuro']
```

STEP 5 – Run your regression and check the results
SEE we have the same results : Slide 24 (SPSS) and Slide 38 (R)

```
X = sm.add_constant(X)
model = sm.OLS(Y, X).fit()
predictions = model.predict(X)
print_model = model.summary()
print(print_model)
```

C:\Users\ukonus1\AppData\Local\Programs\Python\Python39\lib\site-packages\statsmodels\tsa\tsaat
x = pd.concat(x[::order], 1)

OLS Regression Results

Dep. Variable:	TotalSalesEuro	R-squared:	0.679
Model:	OLS	Adj. R-squared:	0.654
Method:	Least Squares	F-statistic:	27.14
Date:	Tue, 28 Sep 2021	Prob (F-statistic):	3.82e-17
Time:	15:07:30	Log-Likelihood:	-492.06
No. Observations:	84	AIC:	998.1
Df Residuals:	77	BIC:	1015.
Df Model:	6		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	48.3076	27.560	1.753	0.084	-6.572	103.187
Facebook	17.8630	4.847	3.685	0.000	8.211	27.515
BannerAds	1.1308	4.612	0.245	0.807	-8.054	10.315
SEA	0.1676	4.813	0.035	0.972	-9.416	9.751
Emails	2.5132	4.160	0.604	0.548	-5.771	10.797
Loyalty_Regular_Dummy1	62.9235	22.584	2.786	0.007	17.953	107.895
Loyalty_Golden_Dummy2	277.7245	33.240	8.355	0.000	211.534	343.915

Regression with Python – Pandas & Statsmodels*



Now let's practice this time how it works for our **Logistic Regression data (Data B - online attribution)** as we used in SPSS/R.

Our libraries (panda & statsmodels) are already imported in previous exercise
So let's start by reading our data and defining our variables right away !

STEP 1 – Import and read your data set (here from spss file)

STEP 2 - Define your X (explanatory) and Y (Outcome) variables in your model

```
>>> df = pd.read_spss('B_Attribution_Logistic_PracticeData.sav')
>>> df.tail()
   Ad1  Ad2  Youtube  Twitter  Facebook  OSearch  SSearch  Conversion
63  1.0  1.0      1.0      1.0       0.0      0.0      1.0          0.0
64  0.0  0.0      0.0      0.0       1.0      1.0      0.0          1.0
65  0.0  1.0      0.0      0.0       1.0      0.0      0.0          1.0
66  0.0  0.0      0.0      0.0       0.0      1.0      0.0          0.0
67  0.0  0.0      0.0      0.0       1.0      0.0      0.0          1.0
```

```
>>> X = df[['Ad1', 'Ad2', 'Youtube', 'Twitter', 'Facebook', 'OSearch', 'SSearch']]
>>> Y = df['Conversion']
```

STEP 3– Run your logistic regression and check the results
SEE we have the same results : Slide 33 (SPSS) and Slide 39 (R)

```
x = pd.concat([x[['Ad1', 'Ad2', 'Youtube', 'Twitter', 'Facebook', 'OSearch', 'SSearch']],
               pd.Series([1] * len(x), index=x.index)), axis=1)
>>> logit_mod = sm.Logit(Y, x)
>>> logit_res = logit_mod.fit()
Optimization terminated successfully.
      Current function value: 0.276571
      Iterations 8
>>> print(logit_res.summary())
```

Logit Regression Results						
=====						
Dep. Variable:	Conversion	No. Observations:	68			
Model:	Logit	Df Residuals:	60			
Method:	MLE	Df Model:	7			
Date:	Tue, 28 Sep 2021	Pseudo R-squ.:	0.6007			
Time:	15:22:09	Log-Likelihood:	-18.807			
converged:	True	LL-Null:	-47.105			
Covariance Type:	nonrobust	LLR p-value:	7.193e-10			
=====						
	coef	std err	z	P> z	[0.025	0.975]

const	-2.8577	1.374	-2.080	0.038	-5.550	-0.165
Ad1	-1.3202	0.969	-1.363	0.173	-3.218	0.578
Ad2	-0.7101	0.999	-0.711	0.477	-2.668	1.248
Youtube	1.1001	0.994	1.107	0.268	-0.848	3.048
Twitter	-0.5716	0.976	-0.586	0.558	-2.485	1.342
Facebook	2.9906	1.337	2.236	0.025	0.370	5.612
OSearch	5.9010	1.423	4.147	0.000	3.112	8.690
SSearch	-0.8730	1.015	-0.860	0.390	-2.862	1.116
=====						

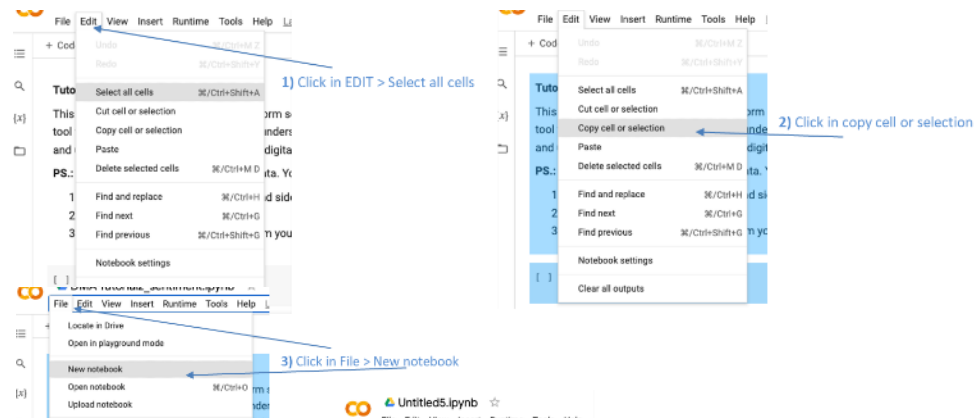
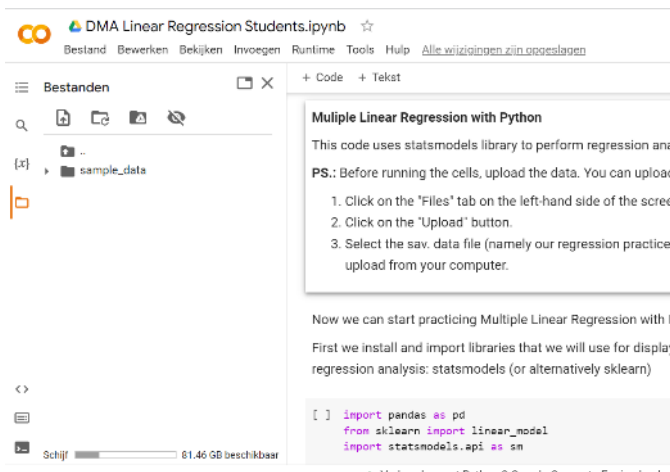
Regression with Python – Pandas & Statsmodels*



OPTIONAL Self Study Exercise*

You can follow the links below to go and repeat the Python exercise on Linear Regression and Logistic Regression yourself at home.

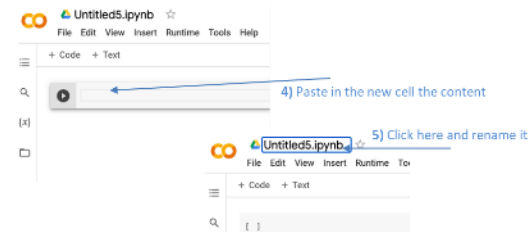
Please copy-paste the all working–file on a new project you create before you run.



PLEASE CLICK THE INTERACTIVE LINKS BELOW TO REACH THE COLAB (PYTHON) NOTEBOOK TO REPEAT OUR EXERCISES WITH PYTHON YOURSELF.

[Link 1: Linear Regression](#)

[Link 2: Logistic Regression](#)



Finally:
Probabilistic Attribution:

Probabilistic Attribution: Value of Each Touchpoint in Journey

conversion rate

Click-path	Frequency	Conversions	CR
A	6000	1	0.0002
B	2000	60	0.0300
C	3000	120	0.0400
AB	350	11	0.0314
AC	300	12	0.0400
BA	250	8	0.0320
BC	200	14	0.0700
CA	150	7	0.0467
CB	100	5	0.0500
ABC	10	1	0.1000
ACB	20	3	0.1500
BAC	30	4	0.1333
BCA	40	3	0.0750
CAB	50	7	0.1400
CBA	60	5	0.0833

PROBABILISTIC ATTRIBUTON: AN EXAMPLE

Let's say we have different journeys where different touchpoints are used (A,B,C)

How do we calculate the contribution (attribution value) of each touchpoint in the journey:

FEW RULES TO KNOW :

1 Position (Sequence) of touchpoint matters in Probabilistic Attribution

2 Thus: we don't only look whether A is used or not used (seen) by customer in journey

3 We also look at its position and use it in our calculation:

I.E Whether A is used in the beginning #1, middle #2 or in the end #3 as the last touchpoint

4 This gives us a more precise/accurate picture on the real effect of each touchpoint

5 Then we know the value of each touchpoint depending on its position in a particular journey

REMEMBER OUR EXAMPLE FROM LECTURE 1:

Let's say we have an CAB journey: where customer first see C, then A and finally B

What is the attribution value of each touchpoint in this path (journey) ?

Probabilistic Attribution: Value of Each Touchpoint in Journey

Click-path	Frequency	Conversions	CR
A	6000	1	0.0002
B	2000	60	0.0300
C	3000	120	0.0400
AB	350	11	0.0314
AC	300	12	0.0400
BA	250	8	0.0320
BC	200	14	0.0700
CA	150	7	0.0467
CB	100	5	0.0500
ABC	10	1	0.1000
ACB	20	3	0.1500
BAC	30	4	0.1333
BCA	40	3	0.0750
CAB	50	7	0.1400
CBA	60	5	0.0833

- $Y = 1$ if conversion or 0 if no conversion
- T_{kj} = touch-point k of click-path j
- S_{ij} = set of the first i touch-points of click-path j , $S_{ij} = \{T_{1j}, T_{2j}, \dots, T_{ij}\}$
- MC_{ij} = marginal contribution of touch-point i to click-path j

$$MC_{ij} = \text{Prob}(Y = 1 | S_{ij}) - \text{Prob}(Y = 1 | S_{i-1,j})$$

- Increase in conversion probability caused by touch-point i for click-path j

DIFFERENCE WITH SIMPLE/RULE BASED TECHNIQUES TECHNIQUES:

1 YOU TAKE PATH: THUS – ORDER OR SEQUENCE OF EACH AD ALONG JOURNEY INTO CONSIDERATION

2 THUS IT IS NOT SIMPLY COUNTING THE AD (TOUCHPOINT) SHARE

2 YOU CALCULATE THE MARGINAL CONTRIBUTION OF EACH TOUCHPOINT (AD) IN JOURNEYS WHERE TOUCHPOINTS (ADS) ARE USED IN DIFFERENT SEQUENTIAL ORDER

SEE THE EXAMPLE IN NEXT SLIDE !

Probabilistic Attribution: Value of Each Touchpoint in Journey

Start with the first one (in sequence, here = C alone) > Go stepwise in the path such as: (CA – C) for A and then (CAB – CA) for B

Very small example with only 3 marketing channels: A, B, C (Note that channel A has little added value to the conversions)

Maybe A is a website that shows our ad, but not to the right audience for our product

Example

- $MC_{C,CAB} = \text{Prob}(Y = 1 | C) - \text{Prob}(Y = 1 | \emptyset)$
 $= 0.04 - 0 = 0.04$
- $MC_{A,CAB} = \text{Prob}(Y = 1 | CA) - \text{Prob}(Y = 1 | C)$
 $= 0.0467 - 0.04 = 0.0067$
- $MC_{B,CAB} = \text{Prob}(Y = 1 | CAB) - \text{Prob}(Y = 1 | CA)$
 $= 0.14 - 0.0467 = 0.0933$
- Conversion Attribution for path CAB:
 - $A = 7 * 0.0067 / 0.14 = 0.33$
 - $B = 7 * 0.0933 / 0.14 = 4.67$
 - $C = 7 * 0.04 / 0.14 = 2.00$

Click-path	Frequency	Conversions	CR
A	6000	1	0.0002
B	2000	60	0.0300
C	3000	120	0.0400
AB	350	11	0.0314
AC	300	12	0.0400
BA	250	8	0.0320
BC	200	14	0.0700
CA	150	7	0.0467
CB	100	5	0.0500
ABC	10	1	0.1000
ACB	20	3	0.1500
BAC	30	4	0.1333
BCA	40	3	0.0750
CAB	50	7	0.1400
CBA	60	5	0.0833

Self Practice: Can you repeat this exercise for BAC and then ABC ? Observe how things change – when touchpoint position (in journey) changes

Appendix

Regression with R (GLM) Library

(Optional: for R Users)

Regression with R – GLM Module

Here is an example with same data (online sales) regression with R by using few lines of R code – which are mostly available in online tutorials.

Suggestion: If you are not familiar with R environment use **R Studio** which is more convenient (has SPSS kindof dashboard)

Morover in R Studio it is easier to import data from external data files in SPSS or Excel format.

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Online_Revenue_Regression_PracticeData				
Filter				
Facebook	BannerAds	SEA		Em
# Facebook: number of Facebook ads clicked in 2017 (X1)	# Banner Ads: number of Banner ads clicked in 2017 (X2)	# SEA: number of Search Engine Advertising (Paid Search) results c...		# E
1	0	2		0
2	0	2		1
3	3	2		4
4	0	2		1
5	0	2		1
6	5	4		4
7	1	4		2
8	1	4		0
9	1	4		1
10	1	4		5
11	1	3		5
12	3	3		6
13	8	3		6
14	1	3		6
15	1	3		6

Showing 1 to 19 of 84 entries

Console Terminal

```
R version 3.5.1 (2018-07-02) -- "Feather Spray"
Copyright (C) 2018 The R Foundation for Statistical Computing
Platform: x86_64-w64-mingw32/x64 (64-bit)
```

```
R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.
```

```
R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.
```

```
Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.
```

```
> library(haven)
warning message:
package 'haven' was built under R version 3.5.3
> online_revenue_regression_practiceData <- read_sav("C:/users/ukonus1/Desktop/Online_Revenue_Regression_PracticeData.sav")
> view(online_revenue_regression_practiceData)
```

Your linear regression SPSS data imported in R Studio

Simple code-line to import data from SPSS to R environment

Note: You can also use "Import" code in the upper panel



Linear Regression with R- GLM Module

Then by using your imported Online Revenue Regression Practice Data, you use and run the linear model: **glm()** R code

You show the results with **summary(fit)** command

Check: You get exactly SAME results that you got in your SPSS exercise (see slides: 29-30-31)

```
R 4.1.0 ~ /  
> fit <- glm(TotalSalesEuro ~ Facebook + BannerAds + SEA + Emails + Loyalty_Regular_Dummy1 + Loyalty_Golden_Dummy2, data=A_Online_Sales_Regression_PracticeData, family=gaussian())  
> summary(fit)
```

Call:
glm(formula = TotalSalesEuro ~ Facebook + BannerAds + SEA + Emails + Loyalty_Regular_Dummy1 + Loyalty_Golden_Dummy2, family = gaussian(), data = A_Online_Sales_Regression_PracticeData)

Deviance Residuals:

Min	1Q	Median	3Q	Max
-165.12	-48.94	-0.47	26.17	514.10

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	48.3076	27.5604	1.753	0.083618 .
Facebook	17.8630	4.8473	3.685	0.000423 ***
BannerAds	1.1308	4.6123	0.245	0.806976
SEA	0.1676	4.8130	0.035	0.972314
Emails	2.5132	4.1600	0.604	0.547539
Loyalty_Regular_Dummy1	62.9235	22.5842	2.786	0.006712 **
Loyalty_Golden_Dummy2	277.7245	33.2404	8.355	2.1e-12 ***

You can change the regression type by changing (regression) Family Type in R-GLM

Here: **family=gaussian()** ; means linear regression
If you change family type to **family=binomial()** ; R will run logistic regression

Logistic Regression with R – GLM Module

Exactly the same procedure. Import this time your Attribution practice data

glm() R code by using binomial() “**logistic regression**” choice option in the end > you show the results with **summary(fit)** command

SAME results that you got in your SPSS logistic regression exercise : (in R this time family=binomial() ; since we have logistic regression)

```
R 4.1.0 ~ /
> fit <- glm(Conversion ~ Ad1 + Ad2 + Youtube + Twitter + Facebook + OSearch + SSearch, data=B_Attribution_Logistic_PracticeData, family=binomial())
> summary(fit)

Call:
glm(formula = Conversion ~ Ad1 + Ad2 + Youtube + Twitter + Facebook +
    OSearch + SSearch, family = binomial(), data = B_Attribution_Logistic_PracticeData)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-2.4859  -0.3183  -0.0596   0.3254   2.3239

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  -2.8577     1.3738  -2.080  0.0375 *
Ad1           -1.3202     0.9685  -1.363  0.1729
Ad2           -0.7101     0.9990  -0.711  0.4772
Youtube        1.1001     0.9938   1.107  0.2683
Twitter       -0.5716     0.9761  -0.586  0.5581
Facebook       2.9906     1.3372   2.236  0.0253 *
OSearch        5.9010     1.4229   4.147 3.37e-05 ***
SSearch       -0.8730     1.0150  -0.860  0.3897
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Please check:
We simply use the same GLM module and accordingly place our Y
and X variables in the model
Only difference since we have LOGISTIC regression this time
Family=binomial()

Thank You

Dr. Umut Konus
u.konus@uva.nl